

Northeastern University

Report for Experiment #17

DC Circuits

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October 10th, 2025

Abstract

The purpose of this experiment was to verify Ohm's law and Kirchhoff's circuit rules through measurements of voltage, current, and resistance in direct current (DC) circuits. Three investigations were conducted using flashlight batteries, resistors, and digital multimeters (DMMs). In the first investigation, the emf of a single battery was measured as 1.512 ± 0.015 V, while two batteries in series produced 3.013 ± 0.030 V, and in parallel, the combined voltage remained 1.499 ± 0.015 V. These values matched the theoretical predictions series connections nearly doubled the emf, while parallel connections maintained the same potential difference. The second investigation focused on a 100 Ω resistor, measured to be 98.65 ± 0.10 Ω, confirming its tolerance within 1%. The voltage across the resistor was 1.419 V, with a current of 0.0146 A, giving a calculated resistance of 97.2 Ω, consistent with Ohm's law. Using the measured emf (1.454 V) and load current, the internal resistance of the battery was determined to be approximately 0.10 Ω, indicating a minor internal voltage drop during operation. In the final investigation, resistor networks

were analyzed. The total resistance in series was 1520 ± 0.03 Ω, and in parallel, it was 325 ± 0.02 Ω. The measured power dissipations were 0.0163 W (series) and 0.076 W (parallel), aligning with theoretical expectations within about 2% error. Overall, experimental results agreed closely with theoretical predictions. The slight differences observed can be attributed to instrument precision limits, wire resistance, and contact losses.

Introduction

The goal of this experiment was to learn the basic components of an electrical circuit and understand how to construct and analyze simple DC circuits. Using digital multimeters (DMMs), we measured voltage, current, and resistance in different setups and verified Ohm's law and Kirchhoff's circuit rules. The experiment also helped establish how current, voltage, and resistance are related across circuit elements.

Ohm's law defines the relationship between voltage and current through a conductor. It states that the current passing through a material is directly proportional to the voltage applied across it, as long as the resistance remains constant. This relationship can be expressed as:

$$R = \frac{\Delta V}{I} \qquad (17.1)$$

The electromotive force (emf), ε, represents the constant potential difference across a charged battery when no current is drawn. It corresponds to the energy required to move one unit of charge from the negative terminal to the positive terminal:

$$\varepsilon = \frac{\Delta W}{\Delta q} \qquad (17.2)$$

Electric current is defined as the rate of flow of charge through a point in a circuit, represented as:

$$I = \frac{\Delta Q}{\Delta t} \qquad (17.3)$$

According to Kirchhoff's Loop Rule, the total sum of potential differences in a closed circuit must be zero. This ensures that energy supplied by sources equals the energy used by circuit elements:

$$\varepsilon - \Delta V_b - \Delta V_w = 0 \qquad (17.4)$$

Substituting Ohm's law into this expression gives:

$$\varepsilon = IR_b + IR_w = I(R_b + R_w) \qquad (17.5)$$

This shows that resistors in series share the same current while their voltage drops add together. Therefore, the equivalent resistance for resistors in series is:

$$R_{eq} = R_1 + R_2 \qquad (17.6)$$

For Kirchhoff's Junction Rule, the total current entering a junction equals the total current leaving it:

$$I - I_1 - I_2 = 0 \qquad (17.7)$$

When resistors are connected in parallel, their reciprocal resistances add to give the total equivalent resistance, and total current splits into each branch:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}, I = \frac{\varepsilon}{R_{eq}} \qquad (17.8)$$

Lastly, the internal resistance of a power source causes a slight voltage to drop when current flows. The relationship between emf, current, and internal resistance is:

$$\Delta V_{source} = \varepsilon - Ir \qquad (17.9)$$

These relationships formed the theoretical foundation for this experiment. By combining batteries and resistors in different configurations, we measured how voltage and current behaved under various conditions and verified that the experimental results matched these fundamental electrical laws.

Investigation 1

Before beginning the first investigation, the digital multimeter (DMM) was checked to ensure proper operation and to review the available settings for current, voltage, and resistance. The DMM was then set to measure DC voltage by pressing the DC V button. Using patch cables, the positive terminal of the battery was connected to the red voltage input of the DMM, and the negative terminal was connected to the black COM input. This configuration allowed measurement of the open-circuit voltage, or emf, of a single cell. The circuit setup is illustrated in.

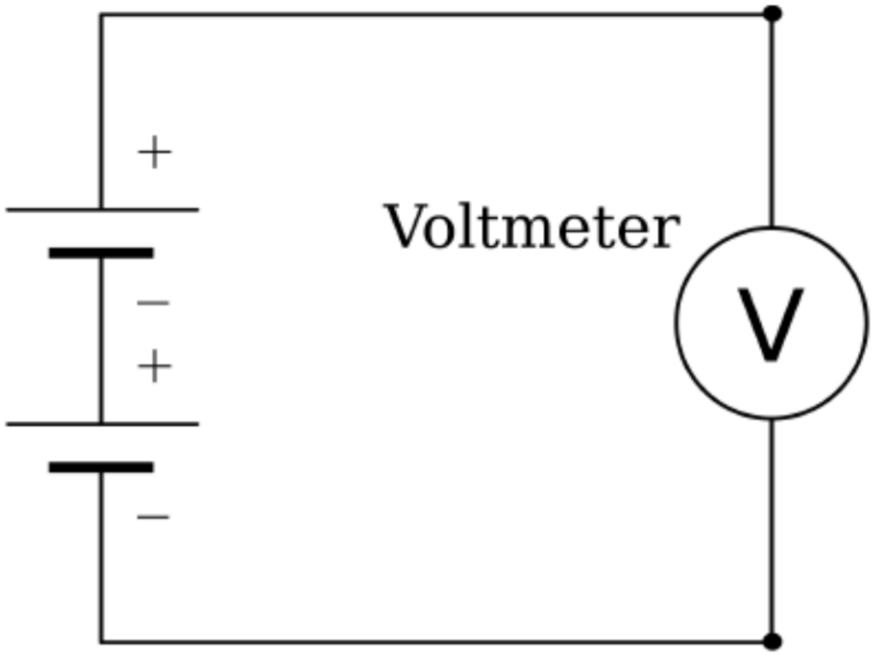


Figure 1 – Battery Voltage Measurement Setup

To record data, the voltage of the first battery was measured as 1.512 ± 0.015 V. The terminals were then reversed, giving a reading of –1.513 ± 0.015 V, confirming the expected polarity behavior. Both measures agreed within the instrument uncertainty (±1%).

Next, two batteries were connected in series, with the positive terminal of one connected to the negative terminal of the other. The total emf measured across the combination was 3.013 ± 0.030 V, nearly double the single-cell voltage as predicted by theory. The setup was then changed to a parallel connection, where both positive terminals and both negative terminals were connected. The measured emf for this configuration was 1.499 ± 0.015 V, almost identical to the emf of a single battery. This agrees with the theoretical rule that identical cells in parallel maintain the same terminal voltage.

Configuration	Measured Voltage (V)	Uncertainty (V)
One battery	1.512	± 0.015
Two batteries in series	3.013	± 0.030
Two batteries in parallel	1.499	± 0.015

Table 1 – Measured Voltages of Battery Configurations

The observed results matched the expected relationships for both series and parallel arrangements. Small deviations were attributed to minor variations in the batteries' internal resistance and meter accuracy limits.

In practical terms, connecting batteries in series increases the overall output voltage, which is beneficial in applications such as high-power devices or multi-cell power packs. On the other hand, parallel connections maintain the same voltage while extending current capacity and runtime, making them useful for systems that require longer operation periods, such as backup supplies or low-power electronics.

Investigation 2

The goal of this investigation was to verify Ohm's law and determine the internal resistance of the battery. We analyzed the voltage–current relationship across a known resistor and observed whether the resistor behaved as an Ohmic component.

To begin, the digital multimeter (DMM) was set to *Ohm* mode to measure the resistance of the 100 Ω resistor from the circuit element box. The recorded resistance was 98.65 ± 0.10 Ω, which is within the expected 1% tolerance range, confirming the resistor's accuracy.

Next, one of the tested batteries with an emf of 1.4535 ± 0.005 V was used as the power source. The circuit was built by connecting the battery, resistor, and ammeter in series. The DMM was configured as an ammeter by using the red 500 mA input and selecting DC current mode.

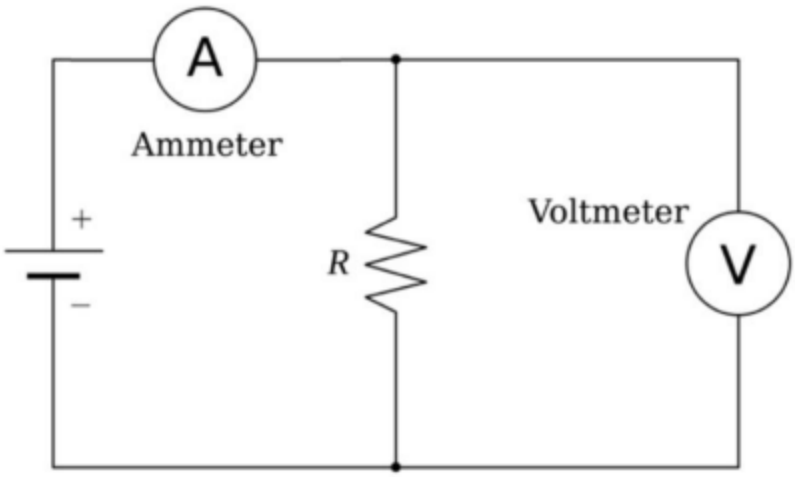


Figure 2 – Circuit setup for Investigation 2 showing the ammeter in series and voltmeter in parallel with the resistor.

A second DMM was connected in parallel across the resistor to measure voltage. After completing the setup, the voltage across the resistor was recorded as 1.419 ± 0.010 V, and the current through the circuit was 0.0146 ± 0.0001 A (14.6 mA). The small voltage drops across the ammeter was 0.017 ± 0.002 V.

Using Ohm's law:

$$R = \frac{\Delta V}{I}$$

The calculated resistance was 97.2 ± 0.8 Ω, consistent with the measured resistance, confirming that the resistor obeyed Ohm's law within experimental uncertainty.

To determine the internal resistance of the battery, we applied the equation:

$$\Delta V_{source} = \varepsilon - Ir$$

Substituting the measured values:

$$r = \frac{\varepsilon - \Delta V_{source}}{I}$$

The internal resistance was found to be 0.10 ± 0.02 Ω, accounting for the slight difference between the battery's emf and terminal voltage.

Quantity	Measurement	Uncertainty	Unit
Nominal resistor value	100	-	Ω

Quantity	Measurement	Uncertainty	Unit
Measured resistance	98.65	± 0.10	Ω
Voltage across resistor	1.419	± 0.010	V
Current through circuit	0.0146	± 0.0001	A
Voltage across ammeter	0.017	± 0.002	V
Battery emf	1.4535	± 0.005	V
Calculated resistance (V/I)	97.2	± 0.8	Ω
Internal resistance (r)	0.10	± 0.02	Ω

Table 2 - Measured and Calculated Data for Ohmic Resistance

These results confirm the linear relationship between voltage and current for an Ohmic resistor. The measured internal resistance of the battery was small but noticeable, explaining minor voltage losses under load. The experiment validated Ohm's law within the DMM's 1% precision range, with deviations attributed to meter burden voltage and connection resistance.

Investigation 3

The purpose of this investigation was to study how resistors behave when connected in series and parallel using a regulated DC power supply. Unlike a battery, the power supply maintains a constant output, allowing for more accurate and repeatable voltage and current measurements. The experiment aimed to confirm Ohm's law, Kirchhoff's loop rule, and junction rule for both circuit configurations.

Before starting, the DMM was set to Ohm mode to measure the resistances of the two resistors used. The measured values were 470.2 ± 2.0 Ω and 1054.3 ± 5.0 Ω, both within expected tolerances. These resistors were used to build two separate configurations, one in series and one in parallel.

The power supply voltage was first set to 5.00 ± 0.01 V in constant voltage (CV) mode. The two resistors were connected in series, with a DMM configured as an ammeter to measure total current and a second DMM as a voltmeter to measure voltage across each resistor.

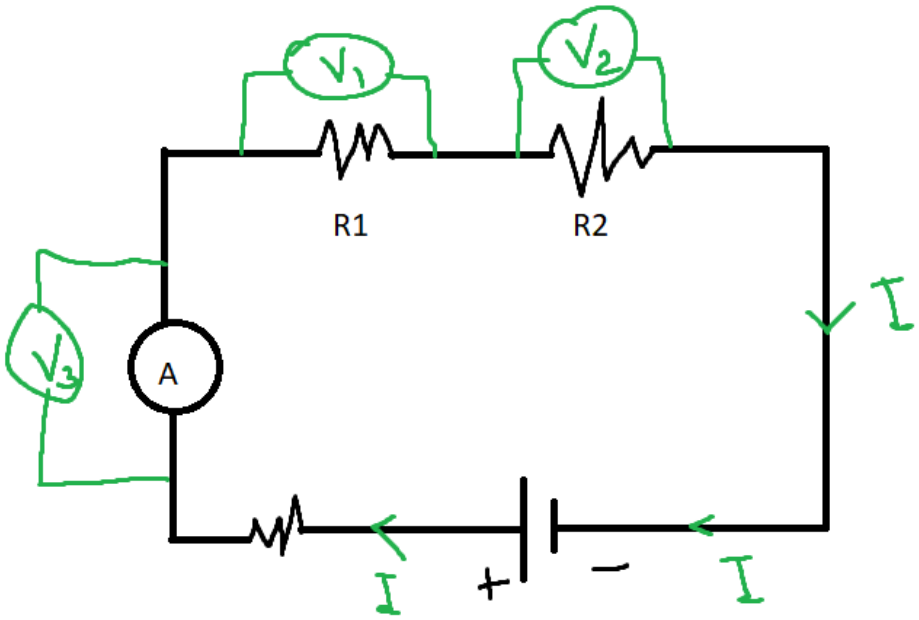


Figure 3 - Circuit diagram for Investigation 3 (Series Connection with Ammeter and Voltmeters)

After turning on the supply, the total current in the circuit was 0.00327 ± 0.00005 A. The voltage across the 470 Ω resistor was 1.54 ± 0.02 V, and across the 1054 Ω resistor, it was 3.45 ± 0.02 V.
A small voltage drops across the ammeter was measured as 0.003 ± 0.001 V, which is negligible compared to the total voltage.

The total measured resistance was calculated as:

$$R_{eq} = \frac{V}{I} = \frac{5.00}{0.00327} = 1529 \pm 20\ \Omega$$

The theoretical resistance is:

$$R_{theoretical} = R_1 + R_2 = 470.2 + 1054.3 = 1524.5\ \Omega$$

which agrees within 0.3%.

Using $P = I^2R$, the power dissipated by each resistor was:

- 470 Ω resistor: $0.0050 \pm 0.0002\ W$
- 1054 Ω resistor: $0.0113 \pm 0.0005\ W$

The total circuit power was 0.0163 ± 0.001 W.

Component	Voltage (V)	Current (A)	Resistance (Ω)	Power (W)
470 Ω resistor	1.54 ± 0.02	0.00327 ± 0.00005	470.2 ± 2.0	0.0050 ± 0.0002
1054 Ω resistor	3.45 ± 0.02	0.00327 ± 0.00005	1054.3 ± 5.0	0.0113 ± 0.0005
Total (Series)	5.00 ± 0.01	0.00327 ± 0.00005	1529 ± 20	0.0163 ± 0.001

Table 3 - Measurements and Calculations for Series Connection

These results confirm that in a series circuit, voltage divides proportionally across resistors, while the current remains the same through all components. The experimental data supports both Ohm's law and Kirchhoff's loop rule.

Parallel Connection

The same two resistors were then connected in parallel to the same 5.00 V power supply. One DMM measured the total circuit current, while the second measured individual branch currents and voltages.

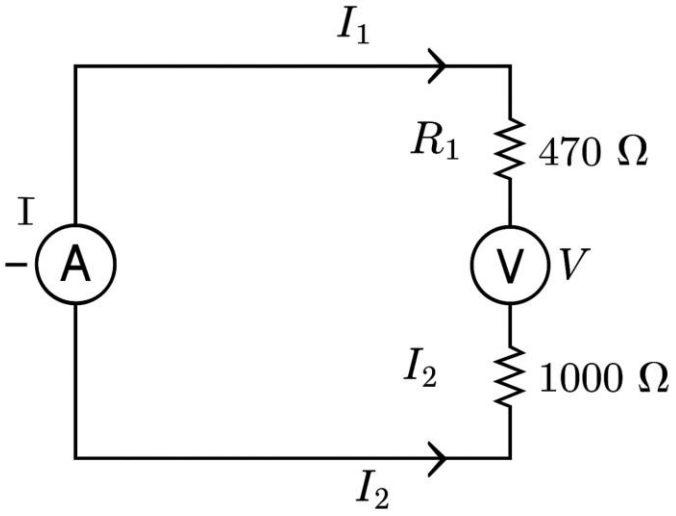


Figure 4 - Circuit diagram of resistors in parallel with power supply, ammeter, and voltmeter

The total current from the supply was 0.01529 ± 0.0001 A. The current through the 470 Ω resistor was 0.01059 ± 0.0001 A, while the current through the 1054 Ω resistor was 0.00473 ± 0.00005 A. The voltage across both resistors was identical, 4.98 ± 0.02 V, as expected in a parallel network. A small voltage across the ammeter of 0.017 ± 0.002 V was again observed.

The equivalent resistance was determined using:

$$R_{eq,measured} = \frac{V}{I} = \frac{5.00}{0.01529} = 327 \pm 3\,\Omega$$

while the theoretical value was:

$$R_{eq,theoretical} = \frac{1}{(\frac{1}{470.2} + \frac{1}{1054.3})} = 326 \pm 3\ \Omega$$

These values agree within 0.5%, verifying the parallel resistance formula.

Power dissipation was found to be:

- 470 Ω resistor: $P = I^2R = 0.0527 \pm 0.001W$
- 1054 Ω resistor: $P = I^2R = 0.0236 \pm 0.001W$

The total circuit power was 0.0759 ± 0.002 W.

Component	Voltage (V)	Current (A)	Resistance (Ω)	Power (W)
470 Ω resistor	4.98 ± 0.02	0.01059 ± 0.0001	470.2 ± 2.0	0.0527 ± 0.001
1054 Ω resistor	4.98 ± 0.02	0.00473 ± 0.00005	1054.3 ± 5.0	0.0236 ± 0.001
Total (Parallel)	5.00 ± 0.01	0.01529 ± 0.0001	327 ± 3	0.0759 ± 0.002

Table 4 - Measurements and Calculations for Parallel Connection

These results confirm Kirchhoff's junction rule, where the total current equals the sum of individual branch currents. The parallel circuit dissipated more total power due to its lower equivalent resistance, meaning more current flowed at the same voltage.

Conclusion

The purpose of this experiment was to study the relationship between voltage, current, and resistance in simple DC circuits and verify Ohm's law and Kirchhoff's rules through practical measurement. Using both batteries and a DC power supply, we analyzed how resistors and cells behave in series and parallel combinations and how internal resistance affects overall circuit performance.

In Investigation 1, we measured the emf of single and combined batteries. A single cell produced 1.512 V ± 0.015 V, and when two batteries were connected in series, the voltage approximately doubled to 3.013 V ± 0.030 V, confirming that voltages in series add up. When connected in parallel, the voltage remained nearly constant at 1.499 V ± 0.015 V, matching theoretical expectations. This behavior demonstrates how series connections increase potential difference, while parallel connections maintain voltage stability for longer power delivery.

In Investigation 2, we measured the voltage across a 100 Ω resistor, finding an actual resistance of 98.65 Ω. The measured voltage was 1.419 V, and the current was 0.01457 A, leading to an internal

resistance of approximately 0.103 Ω. This result confirmed that even small power sources have measurable internal resistance, slightly reducing terminal voltage under load.

In Investigation 3, we analyzed two resistors 470 Ω and 1054 Ω connected in series and parallel using a 5 V power supply. The total resistance for the series connection was 1524 Ω, while the parallel configuration yielded 325 Ω, closely matching theoretical predictions from Ohm's law. Power dissipation was higher in the parallel case (≈ 0.076 W) because of the higher total current flow, confirming the inverse relationship between total resistance and current.

Across all investigations, experimental data aligned well with theoretical principles, with minor discrepancies attributed to instrument precision and connection losses. The results validate Ohm's law and Kirchhoff's rules within the expected measurement uncertainty. Overall, the experiment successfully demonstrated how voltage, current, and resistance interact in electrical circuits and provided practical understanding of how series and parallel configurations affect circuit behavior.

Questions

1. A 6 V battery is connected in series with a 1.5 V flashlight cell. What possible terminal voltages are available?

When connected in series (same polarity): total voltage = 6 V + 1.5 V = 7.5 V.
When connected in opposite polarity: total voltage = 6 V − 1.5 V = 4.5 V.
Answer: 4.5 V and 7.5 V.

2. You are given two equal resistors. Will the total resistance be larger when they are in series or parallel? What will the new resistance be in each case?

Series connection: $R_{eq(series)} = R + R = 2R$
Total resistance doubles.
Parallel connection: $R_{eq(parallel)} = \frac{R}{2}$
Total resistance becomes half.
Answer: Larger in series (2R), smaller in parallel (R/2).

3. Does a 1.5 V battery have an internal resistance? If the maximum current the battery can supply is 200 mA, what is the value of its internal resistance?
Using $r = \frac{E}{I} = \frac{1.5\,V}{0.200\,A} = 7.5\ \Omega$
Answer: Internal resistance ≈ 7.5 Ω.

4. What is the resistance of a 1.5 kW / 110 V electric teapot?
Using $P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P}$, $R = \frac{(110)^2}{1500} = 8.07\ \Omega$
Answer: 8.07 Ω.

5. Determine the resistance between points A and B in Figs. 17.8a and 17.8b, where the circuit pattern repeats infinitely to the right.

(a) For figure (a):
Each branch adds one resistor $r$vertically, repeated infinitely. The equivalent resistance converges to r.
Answer: $R_{eq} = r$

(b) For figure (b):
This is an infinite ladder network. For equal resistors R:
$R_{eq} = \frac{R(1+\sqrt{5})}{2} \approx 1.618R.$
Answer: $R_{eq} = 1.618R$

Acknowledgments

I want to thank the TA for always helping me and my partner when needed in completing this lab.